Weighted Beta-mixture updates for BOLD extensions

Formula schematic of the implemented sequential child-specific scoring rule

Parent component after resolved outcomes

$$p_j \mid h \sim \mathrm{Beta}(A_h, B_h), \qquad w_h \geq 0, \qquad \sum_h w_h = 1$$

$$A_h = a_j + y_j + k_h, \qquad B_h = b_j + n_j^{\mathrm{obs}} - y_j + m_h - k_h$$

Hypothetical untreated DLT: z = 1

$$\mathrm{Beta}(A_h + 1, B_h), \qquad \theta_{h1} = \frac{A_h + 1}{A_h + B_h + 1}$$

$$s_{h1} = \frac{\ell_1 q\, \theta_{h1}}{\ell_1 q\, \theta_{h1} + \ell_0 (1 - \theta_{h1})}$$

$$r_{h1} = w_h s_{h1}$$

Hypothetical untreated non-DLT: z = 0

$$\mathrm{Beta}(A_h, B_h + 1), \qquad \theta_{h0} = \frac{A_h}{A_h + B_h + 1}$$

$$s_{h0} = \frac{\ell_0 (1 - \theta_{h0})}{\ell_1 q\, \theta_{h0} + \ell_0 (1 - \theta_{h0})}$$

$$r_{h0} = w_h s_{h0}$$

Normalize globally, then merge identical Beta components

$$\widetilde{w}_{hz} = \frac{r_{hz}}{\sum_{h'} \sum_{z'=0}^{1} r_{h'z'}}, \qquad f_{\mathrm{new}}(p_j) = \sum_h \sum_{z=0} \widetilde{w}_{hz}\, \mathrm{Beta}(p_j; A_h + z, B_h + 1 - z)$$

Repeat for each latent patient; equal-count branches share one component.

Prediction: $L = + \Rightarrow (\ell_1, \ell_0) = (s, 1 - c)$; $L = - \Rightarrow (1 - s, c)$; no prediction $\Rightarrow (1, 1)$.

Pending untreated outcome: $q = 1 - t/W$. Intervention without observed DLT: $q = 1$.

Observed intervention failure: ordinary DLT update $(A_h, B_h) \mapsto (A_h + 1, B_h)$; no new branch.

s: sensitivity; c: specificity; W: DLT window. Prior: $a_j = \mu_j \nu_j$, $b_j = (1 - \mu_j)\nu_j$.